

Connectivity-Aware Coordination of AUV Fleets in Dynamic Underwater Environments

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Abstract—This work investigates distributed coordination strategies for autonomous underwater vehicles (AUVs) tasked with reaching spatially distributed targets while maintaining line-of-sight connectivity in environments with obstacles and dynamic flow fields. Building on an existing multi-agent control and connectivity framework, we implement a partial version of a previously proposed method and adapt it to our simulation setting. To simplify task allocation and highlight the impact of connectivity constraints, we adopt a reduced approach where tasks are assigned based solely on spatial proximity. Our focus is on evaluating how well control barrier function (CBF) based strategies maintain subgroup and global connectivity under realistic underwater conditions. Simulation results demonstrate the effectiveness and limitations of the partial implementation and proximity-based task assignment, offering insights into practical deployment challenges in energy-constrained, communication-sensitive underwater missions.

Index Terms—Underwater Autonomous Vehicles (AUVs), Multi-Robot Coordination, Underwater Communication Networks

I. INTRODUCTION

We are interested in deploying autonomous underwater vehicles (AUVs) for ocean exploration and sampling, particularly in high risk environments such as beneath polar ice, around hydrothermal vents, and in deep ocean regions where conditions are extreme and unpredictable [1], [2]. These environments are characterized by strong external dynamics, limited visibility, rugged terrain, and severe pressure and temperature fluctuations, all of which pose serious challenges for robotic systems [3]–[6]. The tasks often require the sampling system to dispatch vehicles to separate locations for distinct objectives such as data collection, inspection, or environmental mapping. This spatial dispersion introduces additional challenges, especially in terms of energy efficiency and control.

To mitigate mission risk and improve operational effectiveness, it is often advantageous to deploy fleets of AUVs that operate in coordination. Multi robot systems offer broader spatial coverage, built in redundancy, and distributed task execution, which enhance mission resilience and reduce the consequences of individual vehicle failures [7]–[9]. In such missions, AUVs must spread out across the environment to

efficiently complete their assigned tasks [10]. However, this operational requirement introduces significant coordination challenges. Successful collaboration among AUVs depends on maintaining reliable inter-robot communication. While aerial and terrestrial robots commonly use wireless radio or visual signaling, underwater communication depends heavily on acoustic signaling due to its superior penetration in aquatic environments. Despite this advantage, acoustic communication suffers from low bandwidth, high latency, and vulnerability to environmental interference such as multipath reflections, noise, and signal blockage caused by obstacles [11], [12].

Maintaining an unobstructed line of sight communication structure among AUVs is therefore essential for ensuring robust coordination and safe operation [13]. However, the need for direct communication paths often conflicts with the need for spatial dispersion required to accomplish distributed tasks. For example, AUVs may need to explore regions separated by long distances or occluded by terrain features, which can break line of sight and compromise the integrity of the communication network. Balancing the need to maintain connectivity with the goal of maximizing spatial task coverage remains a central challenge in multi-autonomous vehicles operations [14], [15].

Numerous control strategies have been proposed to manage this trade off. One promising class of approaches involves the use of Control Barrier Functions (CBFs), which are designed to enforce system constraints in real time without requiring full trajectory re-planning [16]. CBFs have been widely used for collision avoidance and are now being explored for additional control objectives. For instance, the Minimum Line of Sight Connectivity Constraint Spanning Tree (MLCCST) algorithm constructs a minimally restrictive spanning tree to maintain global and subgroup LOS connectivity while still allowing robots to move toward distributed task goals [17]. This method formulates the coordination problem as a bi-level optimization and leverages CBFs to enforce safety and connectivity constraints throughout execution.

Much of the existing work assumes idealized conditions that do not reflect the complexity of real underwater environments. In practice, AUVs operate in the presence of dynamic flow fields that include ocean currents, turbulence, and vertices [18]. These environmental forces can significantly alter robot trajectories, degrade control accuracy, and accelerate energy

depletion. This is particularly problematic for long duration missions, where AUVs operate with limited energy reserves and often lack the ability to recharge. Control strategies that fail to account for environmental disturbances risk producing inefficient paths and may lead to mission failure due to premature energy exhaustion.

In this work, we investigate the performance of a CBF-based connectivity maintenance strategy under more realistic underwater conditions that include dynamic flow fields. We implement the MLCCST algorithm in a MATLAB based simulation environment where AUVs are modeled as point mass agents influenced by environmental dynamics. Each robot is assigned an individual goal location, relay nodes are dynamically selected to maintain a line of sight spanning tree, and CBFs are used to enforce safety, obstacle avoidance, and connectivity constraints. We incorporate nonuniform flow fields to simulate realistic underwater dynamics and examine the system's behavior in terms of connectivity preservation, task performance, and energy efficiency. Our goal is to evaluate the robustness and limitations of this approach in more realistic scenarios and to explore improvements for adaptive behavior and energy aware coordination in future multi AUV missions.

The remainder of this workshop paper is organized as follows. Section II presents the system models and outlines the method proposed in [17]. Section III describes the implementation of this approach in a simulated environment. Section IV demonstrates its application under external flow disturbances. Finally, Section V discusses the results and concludes the paper.

II. PRELIMINARIES ON SYSTEM MODELING AND CBF-BASED SOLUTION

In this section, we briefly introduce the system modeling and foundational assumptions, adopting the multi-agent dynamics and connectivity framework presented in [17]. This framework models agents as point-mass systems operating in a continuous environment with limited communication range. Each agent is equipped with local sensing capabilities and can compute control actions in real time. As part of our study, we provide a partial implementation of the methods proposed to reproduce key behaviors under our simulation settings in the next section.

For the purpose of this study, we consider a reduced version of the task allocation problem. Tasks are assigned solely based on spatial proximity between agents and task locations, without performing global optimization. This simplification allows us to concentrate on connectivity maintenance and distributed control, while preserving the essential coordination structure.

A. System modeling

Consider a fleet of autonomous underwater vehicles (AUVs) operating within a confined task space containing polyhedral obstacles. The fleet is assigned to reach spatially distributed

target areas while maintaining global and subgroup connectivity throughout the mission.

1) *Robot Dynamics*: Each AUV is modeled as a single-integrator point-mass robot with first-order kinematics, defined as:

$$\dot{x}_i = u_i, \quad x_i \in \mathbb{R}^2, \quad u_i \in \mathbb{R}^2 \quad (1)$$

where x_i and u_i denote the position and control input of robot i , respectively. This simplification is reasonable during high-level coordination and allows easy integration with control barrier functions (CBFs) for safety and connectivity constraints.

2) *Environment and Communication Model*: The underwater environment is modeled as a continuous free space interspersed with static obstacles $C_{\text{obs}} \subset \mathbb{R}^2$. Each obstacle is represented as a rectangular polygon for simplicity. Robots can only communicate if they satisfy the following LOS-based criteria:

a) *Proximity Constraint*: The Euclidean distance between robots i and j must be less than the communication radius R_c .

b) *Occlusion-Free Visibility*: The line segment between x_i and x_j must not intersect any obstacle in C_{obs} .

The above conditions define the LOS communication graph $G_{\text{LOS}} = (V, E_{\text{LOS}})$, where V is the set of all robots and $E_{\text{LOS}} \subseteq V \times V$ contains undirected edges representing feasible communication links.

B. CBF-based solution

1) *Task Allocation and Control Objective*: Task assignment follows a reduced version of the original work [?], where each robot is assigned to the closest available task based solely on spatial proximity:

$$p_i = \arg \min \|p_i - p_{\text{task}}\|$$

Circular movement is introduced once a robot reaches its task. This is modeled as a position update based on polar coordinates:

$$p_i(t) = p_{\text{task}} + R \cdot [\cos(\theta_i(t)), \sin(\theta_i(t))]$$

Where R is the task radius and $\theta_i(t)$ is the angular position of robot i at time t .

Each AUV is assigned one of four predefined task areas. The mission objective is two-fold:

a) *Goal Achievement*: At least one robot from each task group must reach its designated goal area.

b) *LOS Maintenance*: A connected communication structure must be preserved among all robots using LOS constraints.

To facilitate scalable communication, we seek to maintain the minimum required number of communication links via a Minimum Line-of-Sight Connectivity Constraint Spanning Tree (MLCCST), while guiding the fleet in a tree-like structure toward the task areas.

2) *Control Barrier Functions (CBFs)*: We use control barrier functions to define safe and admissible control inputs under the following constraints:

a) *Inter-Robot Safety:*

$$h_{ij}^s(x) = \|x_i - x_j\|^2 - R_s^2 \geq 0 \quad (2)$$

b) *Robot-Obstacle Avoidance:*

$$h_{io}^{\text{obs}}(x) = \|x_i - x_o^{\text{obs}}\|^2 - R_{\text{obs}}^2 \geq 0 \quad (3)$$

c) *Connectivity Maintenance:*

$$h_{ij}^c(x) = R_c^2 - \|x_i - x_j\|^2 \geq 0 \quad (4)$$

d) *LOS Connectivity (Ellipsoidal Approximation):* For every edge $(i, j) \in E_{\text{LOS}}$, we define an ellipsoidal region enclosing the segment from x_i to x_j and ensure obstacles do not lie within this region. The control constraint becomes:

$$\dot{h}_{ij,o}^{\text{los}}(x, u) + \gamma h_{ij,o}^{\text{los}}(x) \geq 0, \quad \forall o \quad (5)$$

3) *Optimization-Based Controller:* At each timestep, the control inputs are computed by solving the following Quadratic Program (QP):

$$u^* = \arg \min_u \sum_{i=1}^N \|u_i - \hat{u}_i\|^2 \quad (6)$$

Subject to:

$$u \in \mathcal{B}_s(x) \cap \mathcal{B}_{\text{obs}}(x) \cap \mathcal{B}_{\text{LOS}}(x) \quad (7)$$

$$\|u_i\| \leq u_{\text{max}}, \quad \forall i \quad (8)$$

Here, $\hat{u}_i$ represents the nominal task-following controller for robot i . The barrier sets ensure inter-robot safety, obstacle avoidance, and connectivity.

III. IMPLEMENTATION IN A SIMULATED ENVIRONMENT

To evaluate the CBF-based coordination approach, we implemented it in a MATLAB-based simulation environment that models the motion of a team of AUVs operating in a bounded 2D underwater space. The simulation includes key components such as robot spawning, line-of-sight (LOS) graph construction, control barrier function (CBF) formulation, and quadratic program (QP) based motion planning. The scenario involves a fleet of 40 AUVs initially deployed in a predetermined task space. Task locations are precomputed and assigned by a central mothership [10], with each AUV directed to a specific target based on spatial assignment. After all task locations have been visited, the AUVs return and reorganize into a star-shaped formation near the starting location, completing the mission cycle (See Fig. 2).

A. Simulation Setup

The simulation workspace is defined as a 10×10 meter area containing multiple rectangular static obstacles representing submerged debris or seabed features. Four task zones are prescribed at the corners of the map, representing region-specific exploration or surveillance targets. A group of $N = 40$ robots is deployed from a randomly selected task area, with each robot assigned a goal corresponding to one of the four task zones (See Fig. 1).

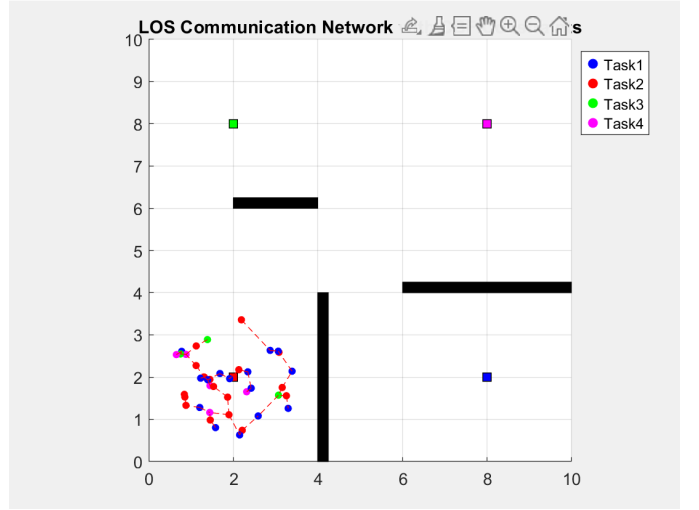


Fig. 1. Initialized simulation taskspace with obstacles and task zones.

B. Robot Control and Connectivity

Robots are modeled using single-integrator dynamics, and their control input is derived by solving a constrained QP at every timestep. The control objectives are to:

- Guide each robot towards its goal area.
- Maintain both global and subgroup LOS connectivity.
- Ensure collision avoidance with other robots and static obstacles.

To ensure LOS-aware communication, a line-of-sight communication graph is dynamically computed using a geometric occlusion check based on ray intersection with polygonal obstacles. A Minimum Volume Enclosing Ellipsoid (MVEE) is also used to approximate and enforce obstacle-free communication regions.

C. Connectivity Maintenance

For each simulation step, a Minimum Line-of-Sight Connectivity Constraint Spanning Tree (MLCCST) is generated using a weight-assignment scheme:

- Edges in the LOS graph are assigned weights based on how close they are to violating safety, connectivity, or LOS constraints.
- A unique weight-penalization strategy is applied to favor edges within the same task group and discourage unnecessary cross-group connections.

The final spanning tree ensures minimum constraint violation while preserving both global and subgroup LOS connectivity.

The MLCCST is built dynamically by solving a Minimum Spanning Tree (MST) for the communication graph that minimizes constraints:

D. LOS Communication Graph:

The communication graph of the AUV fleet is generated such that the graph's adjacency matrix is generated as follows:

$$G_{los}(i, j) = \begin{cases} 1 & \text{if robots } i \text{ and } j \text{ are in LOS range} \\ 0 & \text{otherwise} \end{cases}$$

E. Weight Assignment for Graph Edges:

Weight assignment for graph edges is based on the connectivity constraints, ensuring minimal deviation from nominal behaviors:

$$W_{ij} = -w'_{ij}$$

where w'_{ij} is the computed weight based on constraints.

F. MST Optimization:

To compute the minimal spanning tree:

$$T_w = \arg \min_T \sum_{(i,j) \in T} w'_{ij}$$

G. Control Law Update:

Using Quadratic Programming (QP) for minimal deviation from the nominal task-related controller:

$$u_i^* = \arg \min_u \sum_{i=1}^N \|u_i - \hat{u}_i\|^2$$

H. Task Re-assignment and Removal

Tasks are dynamically assigned and removed after a set time period. Robots adjust their positions accordingly, and new tasks are introduced following the gap period.

Task Re-assignment: After a gap period, robots are re-assigned to new tasks based on proximity:

$$p_i = \arg \min_{p_{task}} \|p_i - p_{task}\|$$

Algorithm 1 Task Generation and Robot Task Assignment

Input:

x - current robot positions $\in \mathbb{R}^{N \times 2}$

Output:

task areas - generated task areas,

robot tasks - assigned tasks for each robot

procedure TASKGENERATIONANDASSIGNMENT(x)

Step 1: Generate new random task locations

Task Areas $\leftarrow$ GenerateRandomTasks(x)

Step 2: Assign robots to closest task

Robot Tasks $\leftarrow$ AssignRobotsToTasks(x , task_areas)

Step 3: Update robot colors based on task assignment
for each robot i do

 Update robot color based on robot_tasks[i]

end for

return Task Areas, Robot Tasks

end procedure

The upgrades to the MLCCST algorithm and the additions for task re-assignment, and formation changes are mathematically formulated to ensure minimal disruption to the task assignments while maintaining connectivity. The integration of Control Barrier Functions (CBFs) for safety, obstacle avoidance, and LOS connectivity ensures real-time robot coordination under dynamic conditions, even when tasks disappear and reappear.

IV. COORDINATION IN FLOW FIELDS

By solving bilevel optimization problems with CBFs at each timestep, the system dynamically adapts to both environmental changes, such as flow forces, and task reassignments. The introduction of external flow fields further enhances the realism of the simulation and demonstrates the framework's robustness under dynamic conditions.

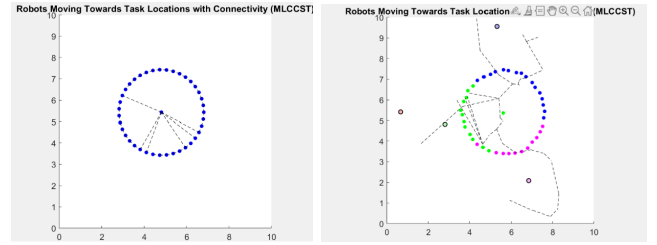


Fig. 2. 40 AUVs starting as a star shape and return to a similar shape at the end of the task.

A. External Flow Force

An external flow force, simulating environmental disturbances such as water current, is applied to the robots:

$$F_{flow}(x_i) = \alpha \cdot (x_{target} - x_i)$$

Where α is a flow coefficient, and x_{target} is the target point (e.g., the central position or goal point).

B. Algorithm

The Updated CBF-based coordination algorithm with external flow force and flow dynamics operates by first computing the communication graph for the robots, where each robot's position is evaluated for connectivity. For each edge in the graph, the algorithm calculates the weights, which are then used to form a Minimal Spanning Tree (MST) to ensure connectivity among the robots. The robots' movement is influenced by a nominal control signal and modified using a flow force, which is applied dynamically based on external environmental factors, such as ocean currents. The robots' positions are updated at each time step, with constraints applied to maintain connectivity and prevent collisions. Additionally, once tasks are completed, the robots return to their original star-shaped formation, adjusting their positions based on the task assignment. This algorithm ensures that robots maintain connectivity while adapting to external forces and moving in coordination to complete tasks (See Algorithm 2).

Algorithm 2 Updated MLCCST with External Flow Force and Flow Dynamics

0: **Input:**

- x : current robot positions $\in \mathbb{R}^{N \times 2}$
- $\hat{u}$: nominal task-related controller $\in \mathbb{R}^{N \times 2}$
- u_{ext} : external flow force $\in \mathbb{R}^{N \times 2}$
- F_{flow} : flow dynamics impact on robots

0: **Output:** u^* : desired minimally modified controller $\in \mathbb{R}^{N \times 2}$

0: **procedure** MLCCST($x, \hat{u}, C_{obs}, u_{ext}, F_{flow}$)

0: **for** each time step t **do**

0: **Step 1:** Compute LOS communication graph $G^{los} = (V, E^{los})$

0: **Step 2:** For each $(v_i, v_j) \in E^{los}$, compute weight: $W_{i,j} \leftarrow -w'_{i,j}$

0: **Step 3:** Generate weighted graph: $G^{los'} = (V, E^{los'}, W')$

0: **Step 4:** Solve for MST:

$$T'_w = \arg \min_{T_w} \sum_{(v_i, v_j) \in \tau} w'_{i,j}$$

0: **Step 5:** Solve modified control law:

$$u^* = \arg \min_{u \in \mathcal{B}^s(x) \cap \mathcal{B}^s(x, x_{obs}) \cap \mathcal{B}^{los}(x, T'_w)} \sum_{i=1}^N \|u_i - \hat{u}_i\|^2$$

0: **Step 6:** Apply input constraint: $\|u_i\| \leq u_{max}, \quad \forall i$

0: **Step 7:** Add flow dynamics:

$$u_i \leftarrow u_i + u_{ext} + F_{flow}(x_i)$$

0: where $F_{flow}(x_i) = \alpha(x_{target} - x_i)$

0: **Step 8:** Update position:

$$x_i(t+1) = x_i(t) + u_i \cdot \Delta t$$

0: **Step 9:** Return to star formation:

$$x_i^{star}(t) = x_i^{initial} + r_i \cdot \cos(\theta_i), \quad \theta_i = \frac{2\pi i}{N}$$

0: **end for**

0: **end procedure**=0

C. Dynamic Relay Allocation and Goal Structuring

To prevent all robots from converging into a single point at the goal, a dynamic relay assignment mechanism is used:

- Robots close to their goal location (within a predefined radius) form a circular formation around the task area.
- Remaining robots dynamically allocate themselves as relays based on the LOS edge lengths. If an edge exceeds a threshold (e.g., 0.5 m), the robot is forced to act as a relay and hold position to maintain connectivity.

This results in a tree-like relay structure extending from the start zone to the goal zone.

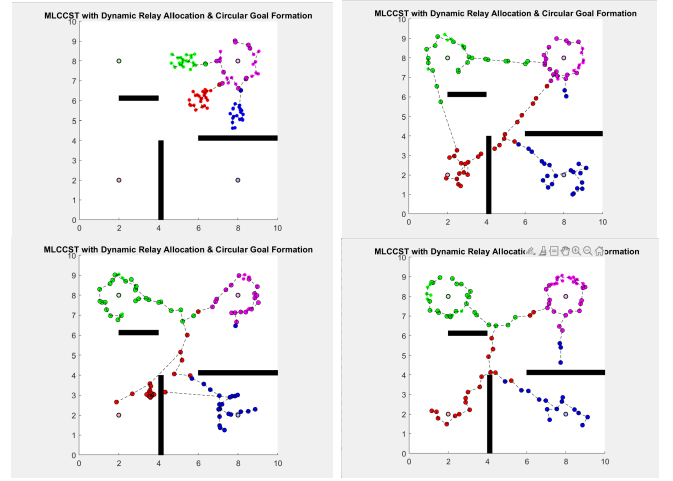


Fig. 3. Simulation output visualizing real-time trajectories of AUVs with color-coded task affiliations. Dashed lines indicate time-varying line-of-sight (LOS) connectivity among robots. Marker size and edge highlights distinguish active robots from relay nodes. The framework ensures all motion and connectivity constraints are maintained while enabling flexible, interpretable, and coordinated multi-agent behavior. See video clip here

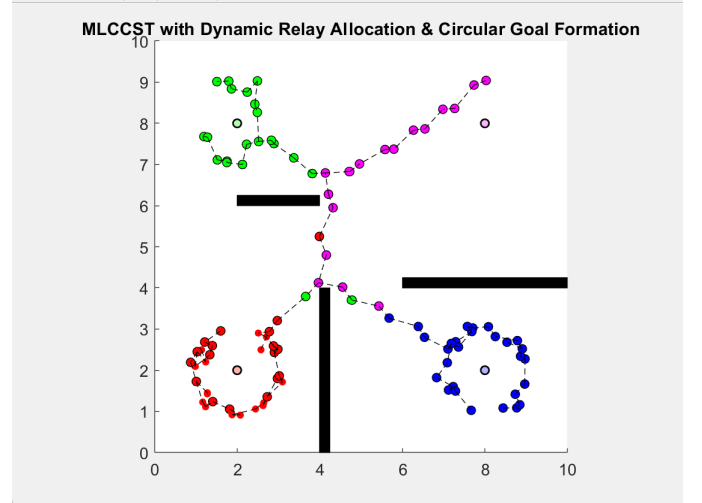


Fig. 4. Final simulation output showing trajectories, task zones, and relay nodes.

D. Summarized results and discussion

This section presents the implementation of the proposed CBF-based coordination framework in simulated underwater scenarios with a fleet of 40 AUVs. Each robot was deployed from a central location and assigned a random goal within four predefined task zones. The framework successfully maintained global and subgroup LOS connectivity while guiding robots toward their goals without violating communication or safety constraints. The resulting formations naturally exhibited tree-like structures, with robots autonomously positioning to relay communication. This behavior emerged from the CBF composition, without requiring explicit coordination rules. The MLCCST algorithm preserved only essential edges for connectivity, enabling scalable and efficient coordination. Sim-

ulations also revealed certain limitations as follows:

- CBF-based obstacle avoidance was less robust in complex or tight environments. Some robots exhibited near-grazing behavior around obstacle edges or became motion-constrained due to conflicting CBF terms.
- In certain cases, excessive convergence was observed at the goal area, causing dense clustering of robots before they spread out into the circular formation.
- Relay nodes occasionally repositioned inefficiently when multiple LOS constraints were simultaneously violated, suggesting the need for more adaptive prioritization.

The results validate the effectiveness of the MLCCST framework in maintaining connectivity and goal-directed motion, even under line-of-sight constraints and external flow fields. By solving bilevel optimization problems and applying control barrier functions at each timestep, the system adapts to environmental disturbances and task reassignments. The inclusion of flow dynamics confirms the method's robustness, while circular movement around task zones helps maintain connectivity without excessive clustering.

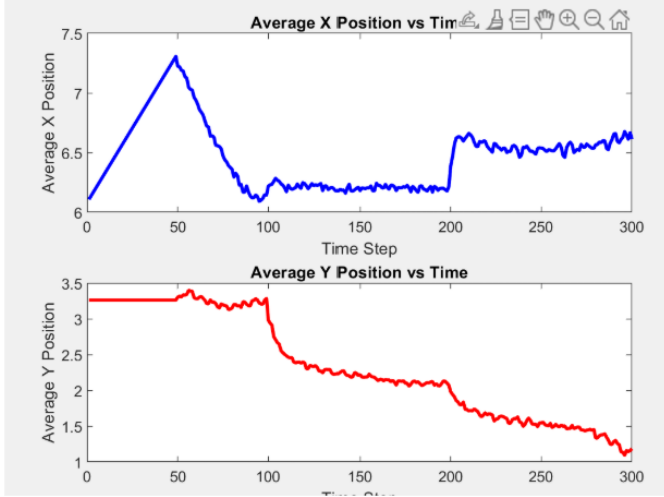


Fig. 5. Average X-axis and Y-axis positions of all AUVs over time

Figure 5 shows the average X and Y positions of the robots over time, capturing their overall movement during the simulation. The sharp initial increase reflects robots moving toward assigned tasks. Once there, circular motion around task centers leads to stabilized averages, especially along the X-axis. After task removal, positions plateau briefly before gradually shifting as robots return to the original star-shaped formation. These trends demonstrate effective task assignment, smooth transitions, and adherence to connectivity constraints under the MLCCST framework.

V. CONCLUSION AND FUTURE WORK

This work implements a CBF-based framework for maintaining LOS connectivity in multi-AUV coordination using the Minimum Line-of-Sight Connectivity Constraint Spanning Tree. AUVs start from a common location and navigate

toward goal areas while preserving both global and subgroup connectivity in environments with obstacles and limited communication. We explore the potential of proposed method in dynamic underwater settings. The simulation, implemented in MATLAB, validates key aspects of the theoretical model, particularly the adaptive assignment of relay robots and the tree-like formation during motion. Robots maintain connectivity using the minimum spanning tree, avoid communication loss through relays, and demonstrate self-organization near goal areas into circular formations.

Simulation results reveal several challenges that remain from the theoretical baseline:

- The CBF-based obstacle avoidance struggles in environments with sharp corners or dense obstacle clusters, occasionally leading to unsafe proximity to obstacles.
- Robots within the same task group often converge on a single goal location, reducing relay diversity and creating redundancy.

Additionally, the current implementation assumes perfect perception of flow fields with no stochastic variation, which does not reflect real-world underwater conditions. To address these limitations and extend the framework, we propose the following future directions:

- 1) *Improved Obstacle Avoidance*: Enhance the CBF formulation for more conservative and robust obstacle avoidance in complex or confined spaces.
- 2) *Dynamic Tree Growth and Shrinkage*: Integrate Control Lyapunov Functions (CLFs) with CBFs to enable adaptive expansion or contraction of the communication tree based on mission demands.
- 3) *Adaptive Goal Reallocation*: Implement dynamic re-allocation within subgroups to optimize task execution and improve relay distribution.
- 4) *Integration with Realistic Underwater Simulation Platforms*: Incorporating realistic hydrodynamics, acoustic delays, and environmental noise in the simulation to bridge toward real-world deployment.

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